

Textural Studies of Retrograded Ductile Shear Zones from Schirmacher Hills, East Antarctica

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Abstract

Mesoscopic and microscopic textural study of discrete ductile shear zones developed in Schirmacher Hills indicate development under low to middle amphibolite facies. Orthomylonites predominate alternating with ultramylonites. Nature of felspar porphyroclasts (σ type) indicates dominance of recrystallisation over rotational deformation during mylonitization. Porphyroclasts as well as matrix aggregates show textures developing during syntectonic growth and post tectonic recovery at elevated temperature. Further, evidence of repeated progressive mylonitization are also preserved in many of these retrograde discrete shear zones.

Introduction and Local Geological Set-up

Schirmacher Hills is a small linear E-W trending range of low lying hill in the Central Dronning Maud Land of East Antarctica. It lies at the interface of continental ice sheet and the shelf ice. Rock exposures are restricted to ice free part of the range which is 17.5 km in length and having maximum width of 2.5 km. During XII Indian Antarctic Expedition, structural studies were carried out in the central part of the hill range between latitudes $70^{\circ}40'-70^{\circ}47'S$ and longitudes $11^{\circ}28'-11^{\circ}45'E$. The area exposes high grade polymetamorphosed ortho and para gneisses (Fig. 1) including biotite-garnet gneiss; augen gneiss; banded gneiss; khondalite, calc. gneiss and migmatites with late intrusion of pegmatites, quartzo-felspathic veins and basic dykes (Sengupta, 1986, 1988; Singh, 1986). The area exposes dominantly charnockite with interlayered mafic granulite layers (emplaced syn D2/2). They contain enclaves of earlier mafic granulites, enderbite and pelitic granulites, recording M1/D1 (Sengupta, 1993). These lithologies have been imprinted by a pervasive retrograde amphibolite facies tectonothermal event (M3/D3) accompanied by syntectonic granitoid emplacement. Further, this amphibolite facies tectonothermal event was accompanied by ductile shearing events, a part of which forms the subject matter

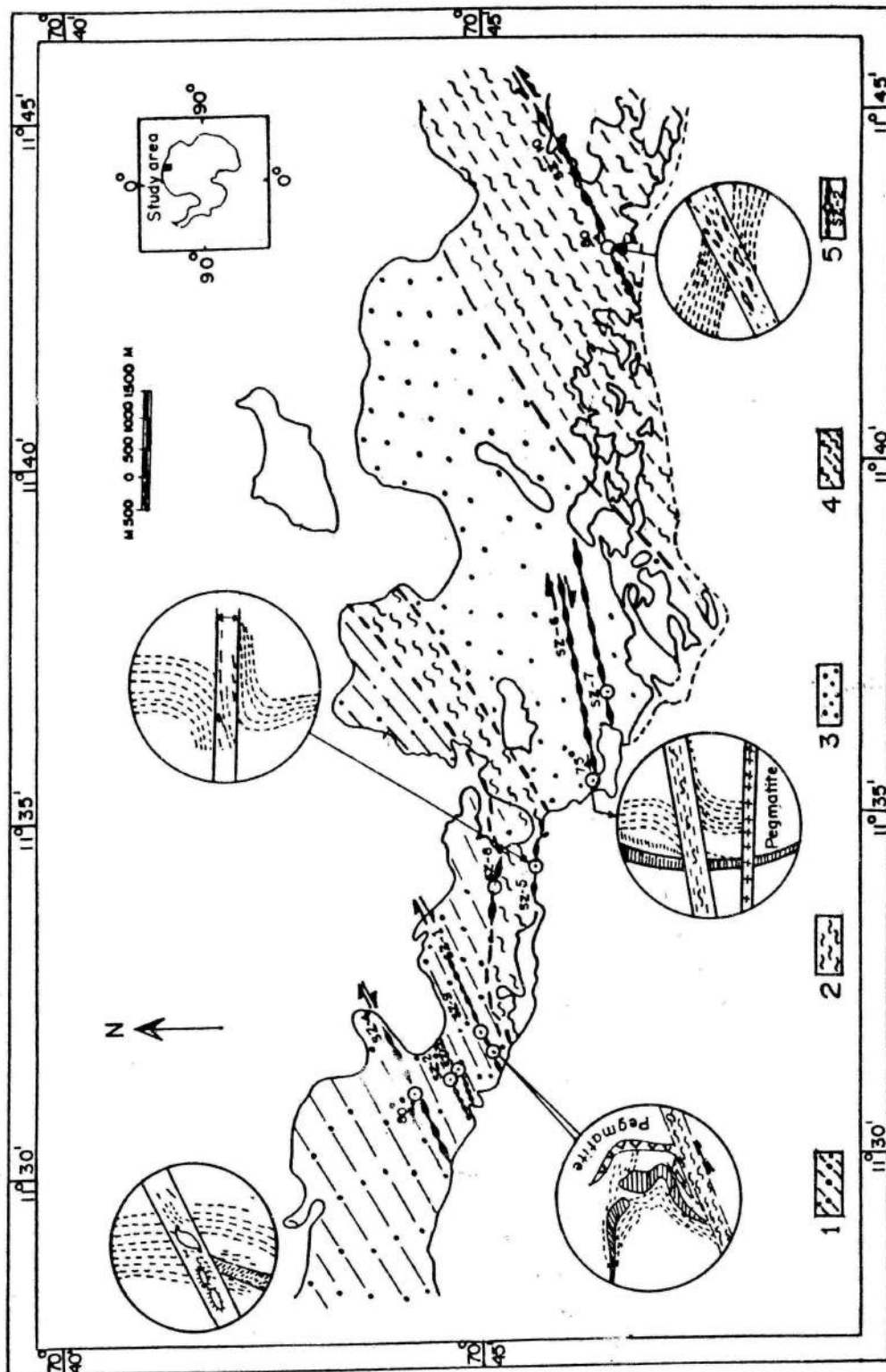


Fig.1. Location of the major shear zones developed in the gneisses: 1. Augen gneiss
2. Khondalite, migmatite and calc gneiss, 3. Garnet-biotite gneiss 4. Banded gneiss
5. Shear zones

of the present study. These are amphibolite grade ductile shear zones, syn M3/D3.

Megascopic Characters of Shear Zones

Three episodes of ductile shearing observed in field effecting the lithounits; early foliation parallel shearing, late steeply dipping striking low to moderate angles to regional foliation and later near N-S striking thin closely spaced shear faults (Sengupta, 1988). Foliation parallel ductile shear zones are essentially S-C mylonites (Lister and Snoke, 1984). Study of the second category discrete ductile shear zones which are at an angle to regional gneissic foliation are presented in Fig. 1. These shear zones have sharp boundaries with gneisses and vary in width from few centimetres to a metre and more and extend laterally sometimes to hundreds of metres. They are roughly parallel to each other with a general trend ENE-WSW to E-W and dip moderate to steep towards south. Though north dipping shears are also noticed. Strike of gneissic foliation which is normally at a high angle to these shear zones, is rotated to become near parallel with the boundary of the shear. Though the average width of these shear zones is about one metre, anastomosing nature is usually seen. These shear zones are characterised by the development of mylonite. Within the shear zones a complete gradation in fabric from protomylonite-orthomylonite (augen mylonite) to ultramylonite has been noticed across the shear zone. Commonly mylonites are layered and show evidences of progressive superposed deformation, *viz.* shear pods, truncated folds etc. At least two mylonitic fabrics (Sma, Smb) can be discerned in most of the mylonites. Stretching lineation defined by elongate quartz and felspar aggregates, usually have low plunges towards WSW. Further few shear zones show steep plunges to southwest and south (Fig. 2). Shear zones showing multiple deformation, have developed two stretching lineations.

Emplacement of pegmatites along these ductile shear zones is observed. Some are clearly synkinematic with respect to the shearing, while others are emplaced at a late stage and show effects of shearing only at their borders or none at all (Fig. 3).

Certain mesoscopic features of these mylonites are presented in Fig. 4 (A-H). It is observed that orthomylonites predominate with preponderance of σ -type porphyroclasts of felspar and garnet (derived from appropriate protoliths). Evidences of repeated mylonitization are seen where the mylonitic fabric is folded and shear lenses/C-surfaces truncate foliation. The bulk sense of shear is determined by deflection of foliation markers into the shear zones (insets in Fig. 1). Further they are corroborated by asymmetric porphyroclasts

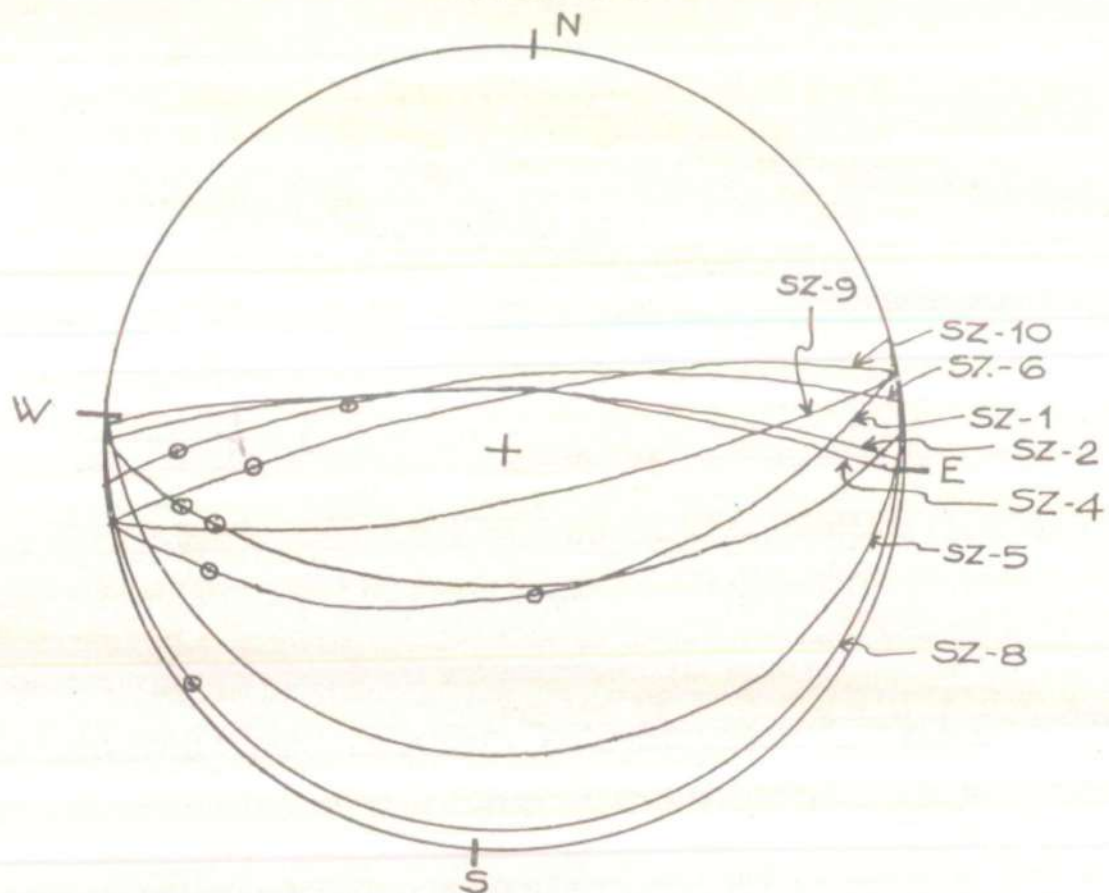


Fig. 2. Stereogram showing shear zones and corresponding stretching lineations.



Fig. 3. Late pegmatite emplaced along the shear zone.

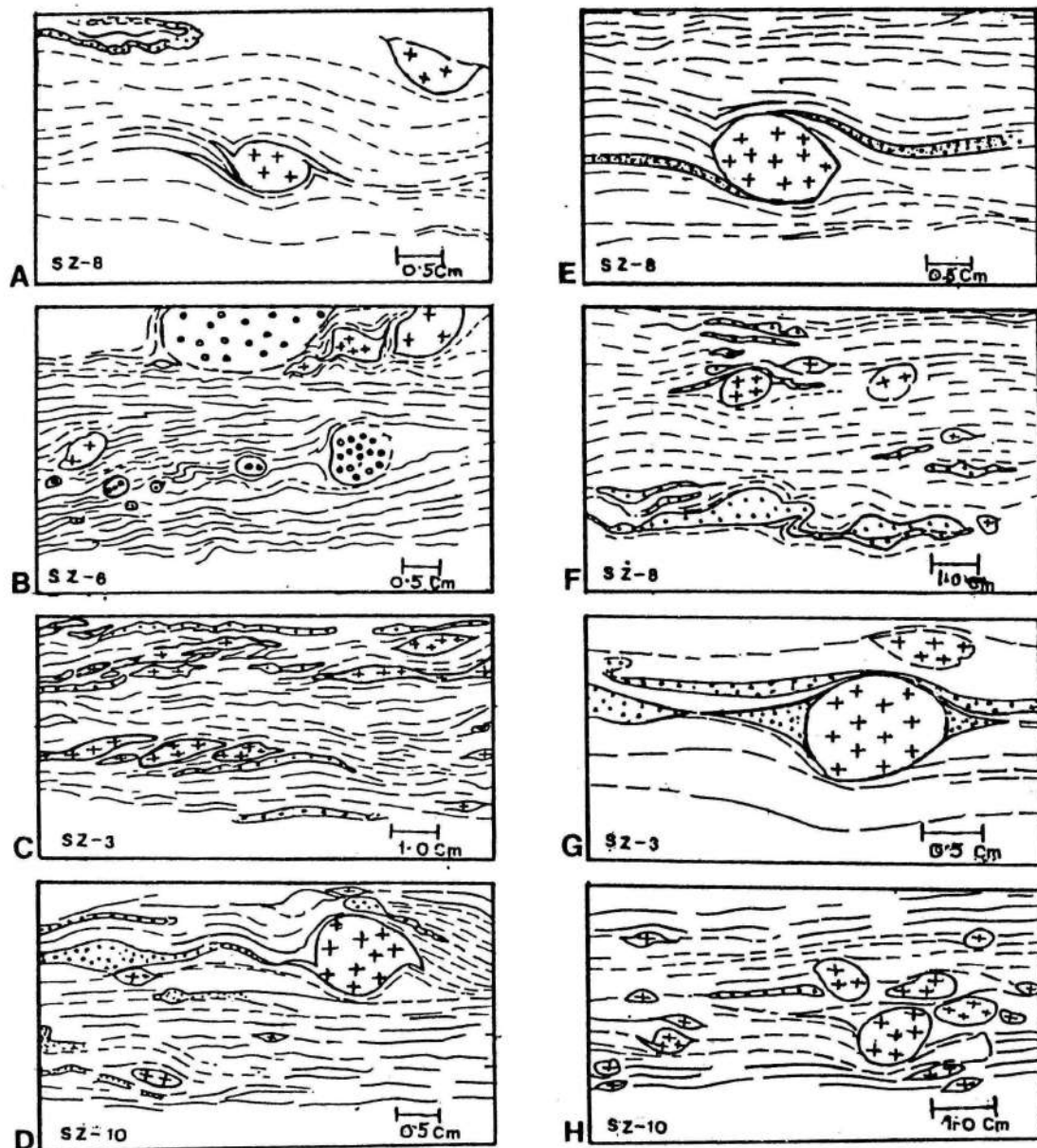


Fig. 4. Symbols used: Plus-Feldspar, Stippled-Quartz, Circles-Garnet, Dashes-Biotite. A. Orthomylonite showing δ -type feldspar porphyroclast, showing sense of shear. Note also tightly folded quartz ribbon, indicating evidence of repeated progressive mylonitization.

B. Garnet and feldspar porphyroclasts showing σ -type geometry. Note foliation parallel shear related folds in matrix. C. Long drawn quartz-ribbons swerving around stacked elongated feldspar porphyroclasts in shear lenses. Orthomylonite.

D. Orthomylonite showing δ -type feldspar porphyroclast showing sense of shear. E. Ultramylonite with few feldspar porphyroclast showing δ -type geometry. Shear sense indicated. F. Orthomylonite showing foliation parallel shearing folding quartz ribbons. G. Near symmetrical tails of quartz around feldspar porphyroclasts. Note quartz-ribbon thins while swerving around porphyroclasts. H. Orthomylonite showing σ -type feldspar porphyroclasts.

in mylonites of felspar and their recrystallised tails (Simpson and Schmidt, 1983; Hanmer and Passchier, 1989).

Majority of the shear zones appear to have developed post dating the F₃ folding event (Sengupta, 1988). Further, metamorphic assemblages (Table 1) preserved within the retrograde shear zones indicate upper greenschist to middle amphibolite facies condition. Emplacement of synshearing pegmatite also indicates broad development of shear zones post dating M₃ granitisation-amphibolite facies event.

Petrographic Characters of Mylonites

General microstructural variations of the quartzofelspathic gneisses in ductile shear zones from Schirmacher Hills have been dealt by Sengupta (1994).

Here, textural aspects of near E-W striking, steeply dipping discrete shear zones, deflecting the regional gneissic foliation are presented (category 2 of Sengupta, 1994). Mylonite terminology is after Wise *et al.* (1984).

As the boundaries of the shear zones are sharp, no gradations from protomylonitic stage were observed; these ductile shear zones are characterised by development of orthomylonites and ultramylonites (Figs. 4 and 6 and Table 1). XZ sections of these mylonites were prepared and their petrography carried out, summarised in Table 1. Fig. 5 shows some typical structures present in these mylonites, including evidences of progressive superposed mylonitization.

As the protoliths of these mylonites are granitic in composition, they are essentially quartzofelspathic mylonites. Early stages of orthomylonites show brittle-ductile deformation of plagioclase (Fig. 7A) and brittle deformation of biotite (Fig. 7B). Well developed banding is seen in them, with alternating layers of quartz ribbons and aggregates of recrystallised quartz-felspar $\pm$ biotite with high angle grain boundaries (Fig. 6A). Grain refinement/recrystallisation of porphyroclasts of felspar are seen (Figs. 6B, 6E, 7E) showing core and mantle structure. These indicate break down of felspar porphyroclasts due to syn-post tectonic growth. Progressive shearing of earlier developed quartz-biotite foliation in orthomylonites are seen by development of discrete C-surfaces (Figs 6C, 6D, 7F). Rotation of felspar and garnet porphyroclasts are indicated by the δ -type porphyroclasts in certain zones which have relatively higher axial ratios and lie oblique to the trace of the foliation. They indicate a higher rate of rotation compared to recrystallisation.

Felspar and garnet porphyroclasts with relatively lower axial ratios are rounded in outline and those with inclusions at angle to external foliation, also

Table 1: Petrography of metamorphic assemblages

Shear Zone	Porphyroclast Type	Porphyroclast Structure	Matrix/Composition and Structure	Assemblage
1. SZ-1	Fs (Mc, Pl), Fs + Qtz lithic, Grt	$\sigma > \delta$	Fine grained quartz, feldspar, Quartz ribbons, biotite. Exhibit core & mantle texture (Fig. 6B) S-C fabric (Fig. 6C).	K-fs+Pl+Qtz+Bt+Grt+Hbl (Zr and Ap as accessories).
2. SZ-2	Mc/Or-perthite, Pl (deformed twin ' mellae), lithic remnants & Grt	$\delta > \sigma$	Fine grained quartz, feldspar, biotite and quartz ribbons. S-C fabrics, brittle fractures in feldspar grains. green Fe rich biotite (retrogressed).	K-fs + Pl + Qtz + Grt Bt + Hbl (altering to Bt & Chl)
3. SZ-3	Fs (Or-perthite), Qz-Fs lithic, Mc, Pl (bent twin lamellae)	$\sigma > \delta$	Fine grained quartz, feldspar, biotite, quartz ribbons. Sub-grains formed at the margin of microcline. Garnet contain inclusions of quartz and biotite. Alteration of garnet to biotite-chlorite quartz.	K-fs + Pl + Qtz + Grt + Bt
4. SZ-4	K-fs-perthite, Pl (deformed twins), Grt, Qtz+Fs aggregates	$\sigma > \delta$	Fine grained quartz, feldspar, biotite (anastomosing), quartz ribbons.	K-fs + Pl + Qtz + Grt+ Bt
5. SZ-6	Grt, Pl, Qz+Fs aggregate	$\delta > \sigma$	Fine grained quartz and feldspar. Shear lenses (Fig. 5C). Relict tight folds (floating hinges).	Grt + K-fs + Pl + Qtz + Bt (Grt retrograding to Bt)
6. SZ-7	Few porphyroclasts of Or-perthite and Pl		Very fine to fine grained quartz, feldspar and biotite. Quartz ribbons.	K-fs+Pl+Qtz+Bt

Contd.

Table 1: Petrography of metamorphic assemblages. – *Contd.*

Shear Zone	Porphyroblast Type	Porphyroblast Structure	Matrix/Composition and Structure	Assemblage
7. SZ-8	Grt, K-fs-perthite, Or		Fine grained felspar, quartz and biotite. Quartz ribbons of two types (earlier ones asymmetrically folded), hinges of folds preserved, recrystallisation around porphyroclasts. Skeletal garnet outlines. Two mylonitic foliation (crenulation of earlier fabric).	Grt+Or+Pl+Bt+Musc+opaques, Biotite with plaeochroic halos (pre-tectonic grains)
8. SZ-9	Grt, K-fs-perthite, Pl	$\sigma > \delta$	Very fine quartz, felspar and biotite. Quartz ribbons, relict tight to isoclinal fold hinges	Qtz+K-fs+Pl+Grt+Bt+ opaques (parallel to foliation)
9. SZ-10	Mc, Or-perthite	$\sigma > \delta$	Fine grained felspar and quartz. Quartz ribbons.	K-fs+Pl+Qtz+Bt+Musc

Fs : Felspar, Or : Orthoclase, Pl : Plagioclase, Mc : Microcline, Qtz : Quartz, Grt : Garnet, Bt : Biotite, Musc : Muscovite, Hbl : Hornblende, Ap : Apatite, Zr : Zircon.

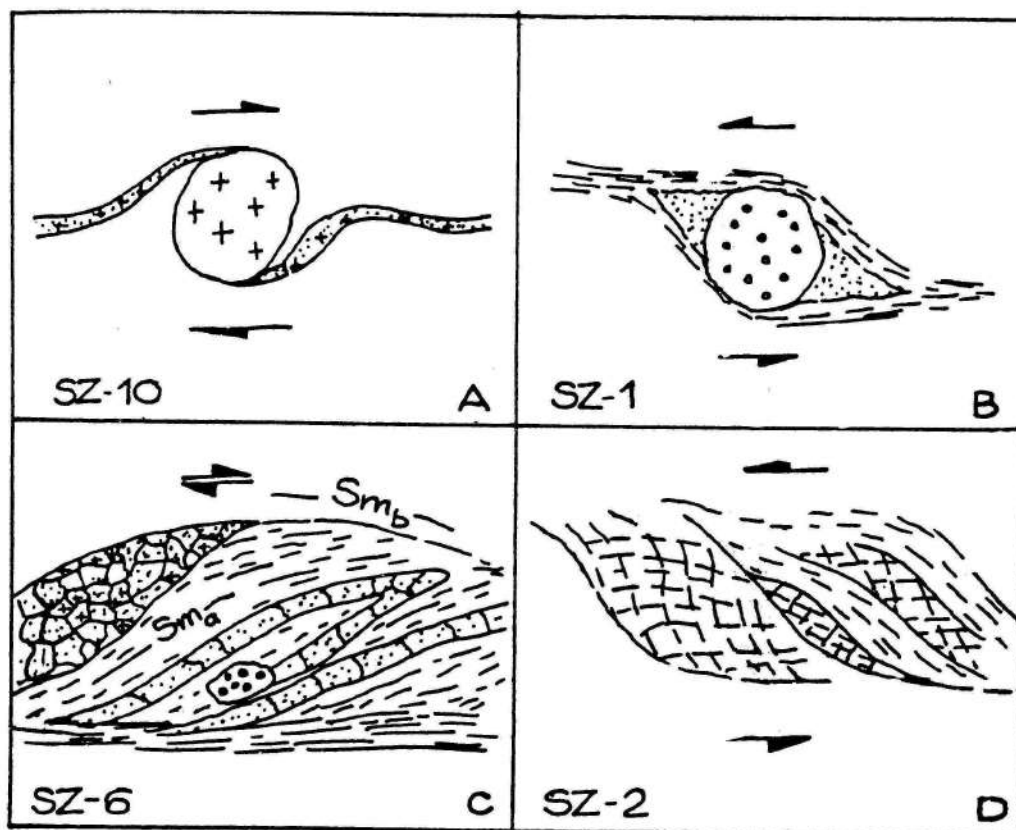


Fig. 5. A. Typical δ -type felspar porphyroblast showing recrystallised tails of quartz-felspar. Geometry indicates dextral shear sense. B. Typical σ -type garnet porphyroblast showing asymmetric quartz tails and curved matrix foliation. Geometry indicates sinistral shear sense. C. Shear lens preserving earlier tight folds on quartz ribbon (Sma). These shear lenses are bound by later mylonitic foliation (Smb). D. Asymmetrical sigmoidal shear lenses of microcline aggregates. Note patterns of deformed cross-hatched twinning by brittle-ductile process indicating pre-tectonic grains deformed by shearing

indicate rotation. Porphyroclasts are dominated by σ -type which indicate a dominance of recrystallisation over rotation during mylonitisation. Quartz porphyroclasts are rare, but occur as aggregates with spindle shaped outlines. Quartz megacrysts essentially occur as ribbons which alternate with recrystallised quartz $\pm$ felspar aggregates. This gives the rock a banded appearance. These quartz ribbons swerve around porphyroclasts and thin at contact with felspar porphyroblast indicating flattening during shearing. Quartz ribbons of single grain have high aspect ratios in XZ sections; otherwise they consist of subgrains in near optical continuity. They are also strain free indicating post-tectonic recovery. This feature in formation of polygonal aggregates is common indicating metamorphism/recovery processes outlasting deformation (Figs. 7C, 7D); these may be termed blástomylonites. Sengupta (1994) has stated that

these quartz megacrysts have developed by grain growth syntectonic to mylonitisation. Large grain size of recrystallised felspar also indicate grain growth post dating cessation of mylonitisation.

The matrix foliation finely anastomoses and consists of aligned biotite, recrystallized quartz and felspar. They usually show serrate to interlobate grain boundaries, deformation bands and undulose extinction, except for undeformed grains which show a polygonal foam texture. Much of the matrix texture represent recovery, with many stages of cyclic processes of grain deformation and recrystallization preserved.

Evidences of repeated progressive mylonitisation seen in mesoscale are distinctly reflected in microscale too. Quartz ribbons and the anastomosing foliation (qz $\pm$ fels - bio) form the major mylonitic foliation. Observations that intrafolial folds on biotite layers and recrystallised quartz ribbons with new biotite flakes/layers and new quartz ribbons axial planar to these folds indicate that this mylonitic foliation has resulted from transposition of an early mylonitic fabric (Figs. 6G, 6H, 7C). Similarly, anastomosing shear lenses superposed on the pre-existing foliation resulted in preservation of the earlier mylonitic fabric (Sma) within micro low strain zones of the superposed mylonitic fabric (Smb) (Fig. 6F).

The deformation is essentially non-coaxial and later reactivation parallel to foliation are seen in development of asymmetric tight folds on mylonitic foliation. That the deformation was progressive is indicated by localization of these transposition structures within the shear zone alone, and are not traceable in the host gneisses.

Chemical Behaviour across a Shear Zone *vis-a-vis* the Host Gneisses

An attempt has been made to study the chemical behaviour of mylonite across the width of a shear zone (SZ- 1) in terms of major oxides as compared to the host rock (gneisses). A comparison of concentration of SiO₂, Al₂O₃, K₂O, Na₂O, CaO, MgO, FeO, TiO₂, and MnO recorded from margin to margin of a shear zone with respect to that of host rock has been presented in Fig. 8. The composition has been expressed in terms of weight percentage. The bulk composition of host rock, *i.e.* augen gneiss shows its granitic affinity. In terms of SiO₂ and Al₂O₃ there is not appreciable change across shear and the host rock; K₂O has considerably increased in the shear zone where increase is relatively higher at the margins as compared to the central part of the shear zone. In contrast Na₂O has recorded a uniform decrease across the shear zone. CaO has behaved in different manner, it has decreased at the margin and slightly increased at the centre whereas MgO has shown continuous depletion except

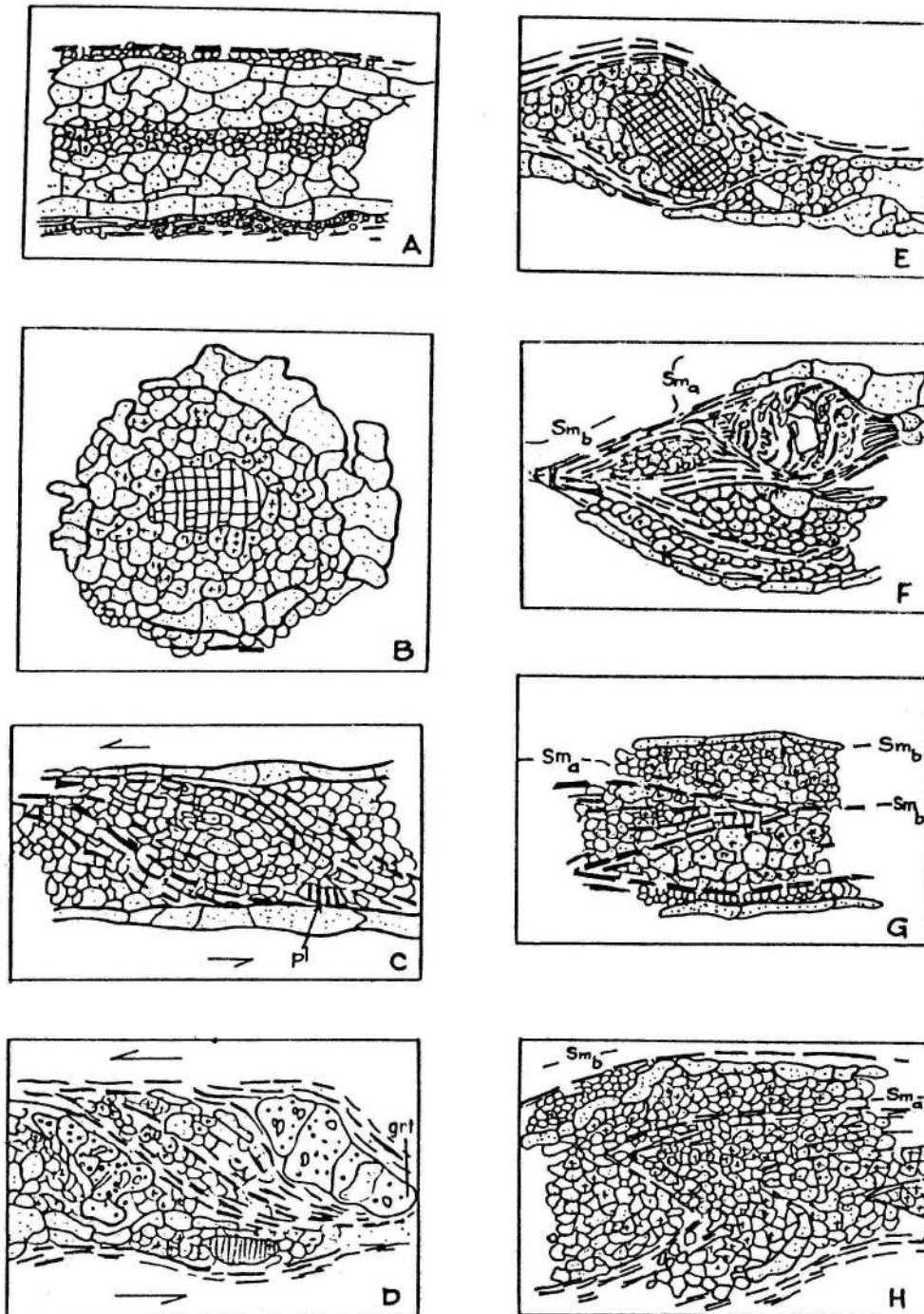


Fig. 6. A. Orthomylonite showing typical development of quartz ribbons alternating with recrystallized matrix aggregates of quartz-felspar-biotite (defining mylonitic foliation). B. Core and mantle structure. Felspar porphyroblast surrounded by recrystallised quartz and felspar aggregates. C. Sigmoidal early mylonitic fabric truncated by later quartz ribbons. Mylonitic fabric defined by recrystallised quartz-felspar-biotite aggregates. D. Sigmoidal fabrics(s) truncated by shear fabric (c). Note s-type garnet porphyroblast. E. σ -type felspar porphyroblast showing later recrystallization to form a core-mantle structure. F. Shear lenses preserving earlier formed mylonitic fabric (S_{ma}) superposed by later anastomosing shears (S_{mb}). G. Tight fold on earlier formed mylonitic foliation (S_{ma}) defined by biotite-quartz-feldspar recrystallized aggregates. Later mylonitic foliation (S_{mb}) defined by quartz ribbons (non-polygonised) and new biotite are axial planar to these folds. H. Early tight folds on mylonitic foliation (S_{ma}) preserved within diffused shear lense with later mylonitic foliation (S_{mb}) swerving around them

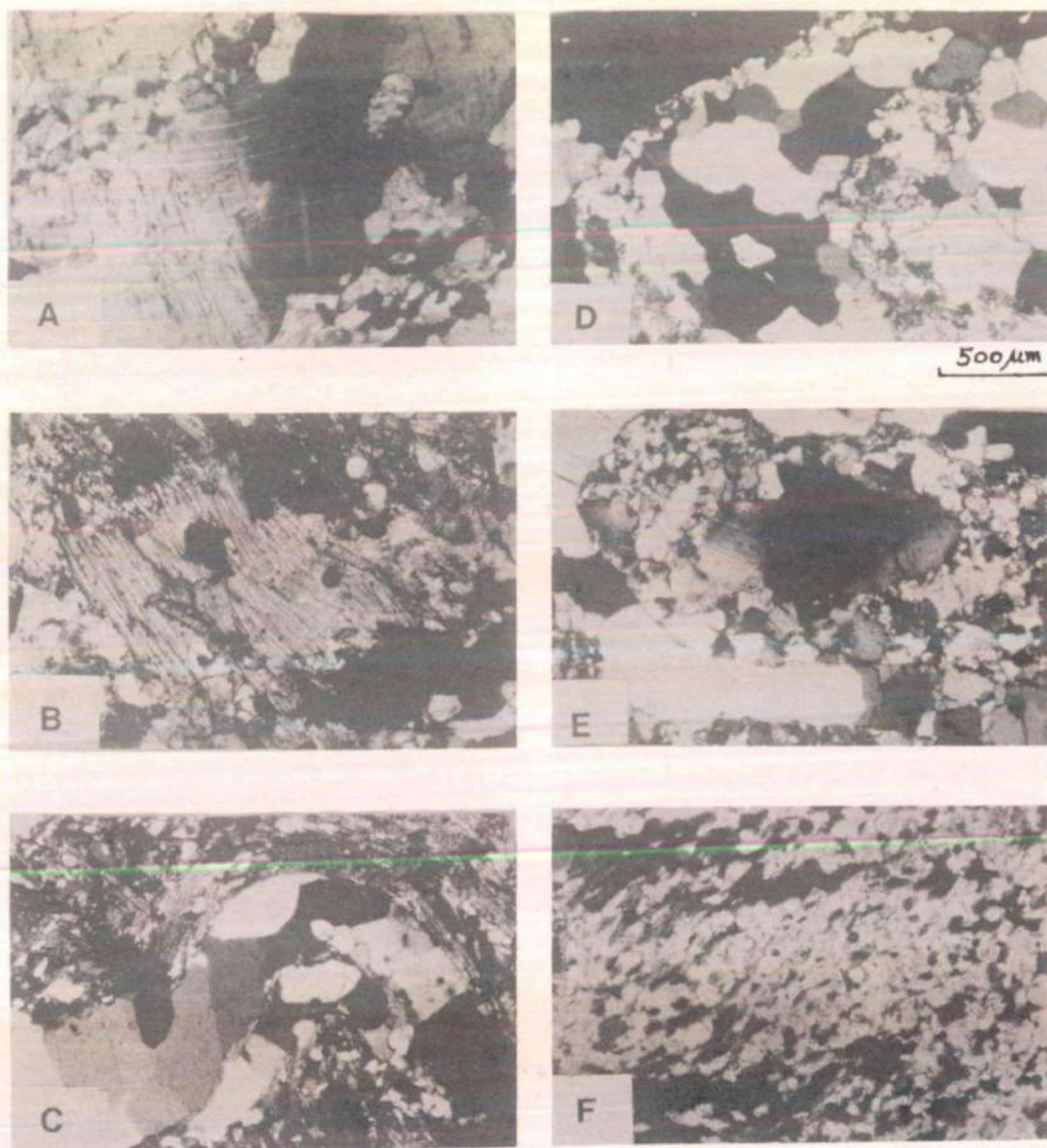


Fig. 7. A. Deformed plagioclase twin lamellae indicating brittle ductile deformation. B. Deformed (kinked) biotite flakes indicating brittle deformation. C. Folded earlier quartz ribbon along with biotite-quartz mylonitic fabric. New quartz-biotite fabric developed axial planar to the fold. Recovery by development of new grains can be seen in large quartz ribbons. D. Folded early quartz ribbons along with quartz-feldspar $\pm$ biotite matrix aggregates. Note similar recovery and recrystallization forming polygonal aggregates in quartz ribbon. E. Core and mantle structure. Porphyroblast of feldspar aggregate (plagioclase-orthoclase) surrounded by neocrystallized quartz and feldspar grains showing polygonal shapes (foam texture). Quartz ribbons swerve around porphyroblast. F. Sigmoidal S-fabric defined by quartz-biotite curving into C-fabric defined by larger biotite aggregates.

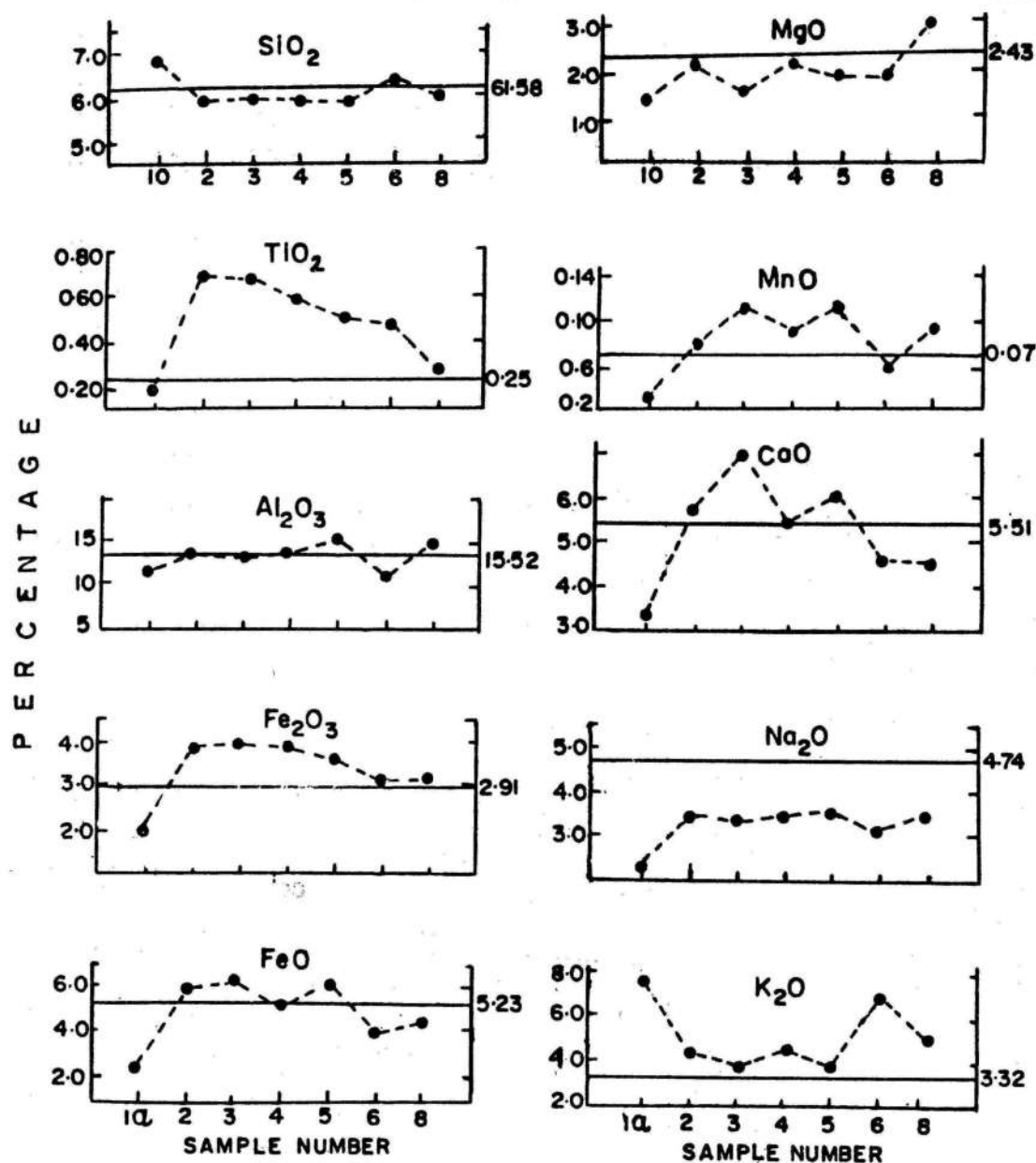


Fig. 8. Chemical variation of major oxides across a shear zone with respect to the external host gneiss (shown as a firm straight line)

at one end. Fe_2O_3 shows slight induction but FeO has remained static. The observations can be explained by the relatively isochemical retrograde (metamorphic) shear zone development except for fluid activity. Increase in K_2O is modally expressed as increased microcline and neo biotite (biotitization). Na_2O appears to be preferentially leached out along with fluids and CaO is just redistributed along with Fe-Mg oxides. This indicates that the fluid activity was important in localisation and development of retrograde mylonitic fabrics. The fluid essentially should have been H_2O (hydration of assemblages).

Conclusions

1. Discrete retrograde ductile shear zones, exhibiting amphibolite facies mineralogy were mapped, which have developed within upper amphibolite-granulite facies gneisses of Schirmacher Hills.
2. These discrete ductile shears appear to have developed in response to continued deformation during uplift and cooling of this crustal section when it became comparatively stiffer.
3. Orthomylonites dominate in shear zones, but they are seen to alternate with ultramylonites. Mylonite fabrics are controlled by host gneiss mineralogy; porphyroclasts are dominated by feldspars and/or garnet with σ -type dominating over δ -type indicating a predominance of recrystallisation over rotational deformation. Banded nature of orthomylonites are due to quartz ribbons alternating with recrystallised quartz-feldspar- biotite aggregates.
4. Many shear zones show evidence of repeated mylonitisation; this has resulted in transposition of early formed mylonitic fabrics.

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